Performance Estimation Problems (PEPs)

Systematic, principled, and computer-aided approaches to the analysis and design of first-order optimization algorithms

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SMAI-MODE tutorial – March 2026

Forewords and problem statement

Numerical examples & postprocessing

Design techniques
- Design via idealized algorithms
- Design via convex relaxations
- Design via brutal approaches

Concluding remarks

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Numerical examples & postprocessing

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Creating new algorithms

A "generic" first-order method

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\begin{aligned}
w_1 &= w_0 - \alpha_{1,0}\nabla f(w_0) \\
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w_3 &= w_2 - \alpha_{3,0}\nabla f(w_0) - \alpha_{3,1}\nabla f(w_1) - \alpha_{3,2}\nabla f(w_2) \\
&\vdots \\
w_N &= w_{N-1} - \sum_{i=0}^{N-1} \alpha_{N,i}\nabla f(w_i),
\end{aligned}
\tag{FOM}
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for some coefficients $\{\alpha_{i,j}\}$. Generic **non-adaptive** first-order method.

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How to choose $\{\alpha_{i,j}\}$?

- pick a performance criterion, for instance $\frac{\|w_N - w_\star\|^2}{\|w_0 - w_\star\|^2}$,
- solve the minimax (minimize worst-case): $\min_{\{\alpha_{i,j}\}_{i,j}} \max_{f\in\mathcal{F},\{w_i\}} \frac{\|w_N - w_\star\|^2}{\|w_0 - w_\star\|^2}$.

Traditional design

Algorithm design (with guarantees):

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Algorithm design (with guarantees):

- unconstrained quadratic programming:

  first-order methods $\Leftrightarrow$ polynomials $\rightarrow$ optimal algorithms via optimal polynomials.[1,2,3]

[1] Golub and Varga (1961). "Chebyshev semi-iterative methods, successive overrelaxation iterative methods, and second order Richardson iterative methods." Numerische Mathematik.

[2] Polyak (1987). "Introduction to optimization." Optimization Software.

[3] Nemirovski (1995). "Information-based complexity of convex programming." (Lecture notes)

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- Beyond quadratics: traditionally more "hand-crafted"
  - Analogy with conjugate gradients.[4,5]
  - Lyapunov function-type analyses.[6,7]

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[4] Nemirovski (1982). "Orth-method for smooth convex optimization." (in Russian) Engineering Cybernetics.

[5] Narkiss and Zibulevsky (2005). "Sequential subspace optimization method for large-scale unconstrained problems."

[6] Nesterov (1983). "A method for solving the convex programming problem with convergence rate $O(1/k^2)$." Dokl. akad. nauk. Sssr 269.

[7] Beck and Teboulle (2009). "A fast iterative shrinkage-thresholding algorithm for linear inverse problems." SIAM journal on imaging sciences 2(1).

Design problem

How to solve the design problem (or proxy of it)?

$$\min_{\{\alpha_{i,j}\}} \max_{f \in \mathcal{F}} \frac{\|w_N - w_\star\|^2}{\|w_0 - w_\star\|^2} ?$$

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- analogies (e.g., with conjugate gradient methods)

$$w_{k+1} \in \operatorname*{argmin}_{x\in\mathbb{R}^d} \{f(w) : \; w \in w_0 + \operatorname{span}\{\nabla f(w_0), \ldots, \nabla f(w_k)\}\},$$

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Recall primal problem, with step-size optimization

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\min_{\alpha_{1,0}} \max_{G,\,F} \quad & G_{1,1} + \alpha_{1,0}^2 G_{2,2} - 2\alpha_{1,0} G_{1,2} \\
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"Simple" minimization problem by dualizing inner maximization.

Dualize inner maximization $\rightarrow$ min min.

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Indeed:

$$\min_{\tau,\lambda \geqslant 0} \quad \tau$$

$$\text{subject to} \quad \begin{bmatrix} \tau - 1 + \frac{\lambda L\mu}{L-\mu} & \alpha_{1,0} - \frac{\lambda(\mu+L)}{2(L-\mu)} \\ \alpha_{1,0} - \frac{\lambda(\mu+L)}{2(L-\mu)} & \frac{\lambda}{L-\mu} - \alpha_{1,0}^2 \end{bmatrix} \succcurlyeq 0.$$

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Optimize $\alpha_{1,0}$ "for free" (linear SDP via Schur complement):

$$\min_{\tau,\lambda\geqslant 0,\alpha_{1,0}} \tau$$

$$\text{subject to} \begin{bmatrix} \tau - 1 + \frac{\lambda L\mu}{L-\mu} & -\frac{\lambda(\mu+L)}{2(L-\mu)} & 1 \\ -\frac{\lambda(\mu+L)}{2(L-\mu)} & \frac{\lambda}{L-\mu} & -\alpha_{1,0} \\ 1 & -\alpha_{1,0} & 1 \end{bmatrix} \succcurlyeq 0.$$

Optimizing the step-sizes ($N = 2$)

When $N = 2$, the problem becomes

$$\min_{\substack{\tau, \lambda_1, \ldots, \lambda_6 \geqslant 0 \\ \{\alpha_{i,j}\}}} \tau$$

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In particular

$$S_{1,2} = -\frac{L\lambda_3 - 2(L-\mu)\alpha_{2,0} + \mu\lambda_1 + L\mu(\lambda_2+\lambda_5)\alpha_{1,0}}{L-\mu}$$

$$S_{2,2} = \frac{-2(\mu\lambda_6 + L\lambda_4)\alpha_{1,0} - 2(L-\mu)\alpha_{2,0}^2 + L\mu(\lambda_2+\lambda_4+\lambda_5+\lambda_6)\alpha_{1,0}^2 + \lambda_1+\lambda_3+\lambda_4+\lambda_6}{L-\mu}$$

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LMI convex in some step-sizes ($\alpha_{2,0}$ and $\alpha_{2,1}$) but not in the others.

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What to expect+what to do with outputs?

Numerical examples I

Example for $L = 1$ and $\mu = .1$

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$$[\alpha^\star_{i,j}] = \begin{bmatrix} 1.8182 \end{bmatrix}.$$

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- For $N = 4$, we reach $\frac{\|w_4 - w_\star\|^2}{\|w_0 - w_\star\|^2} \leqslant 0.0944$ with

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Numerical examples II

What about different performance measure? Example $\frac{f(w_N)-f_\star}{f(w_{\mathbf{0}})-f_\star}$ and $L = 1$, $\mu = .1$.

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- For $N = 4$, we obtain $\frac{f(w_4)-f_\star}{f(w_0)-f_\star} \leqslant 0.0789$ with

$$[\alpha_{i,j}] = \begin{bmatrix} 1.9187 & & & \\ 0.4098 & 2.1746 & & \\ 0.1796 & 0.5147 & 2.1746 & \\ 0.0627 & 0.1796 & 0.4098 & 1.9187 \end{bmatrix}.$$

Numerical example III

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- ◇ worst-case performance of known methods, namely Triple Momentum Method (TMM) and Accelerated/Fast Gradient Method (FGM) computed using PEPs,

Numerical example III

Worst-case performance $\frac{f(w_N)-f_\star}{\|w_0-w_\star\|^2}$ with $L=1$ and $\mu=.01$. We compare

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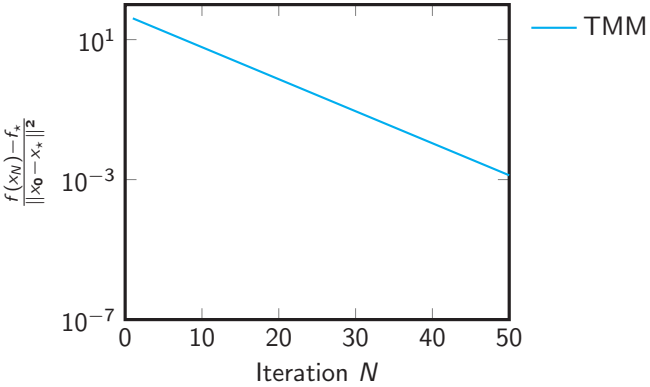

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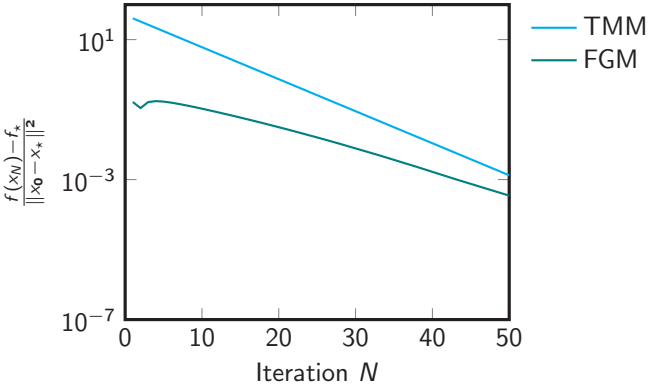

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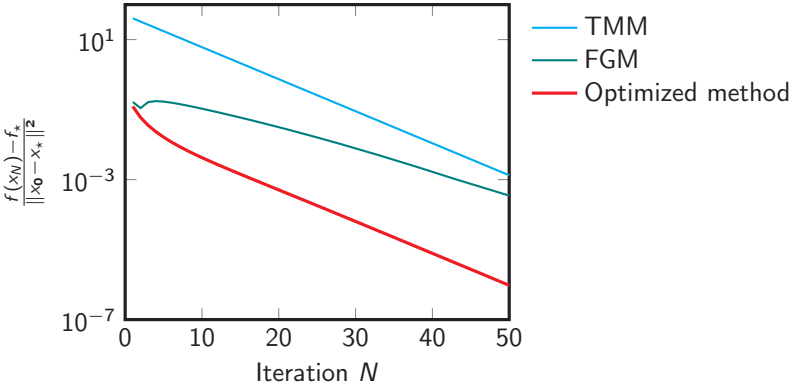

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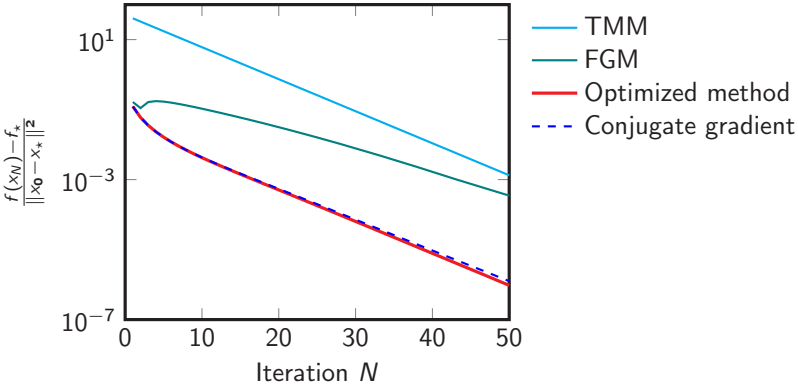

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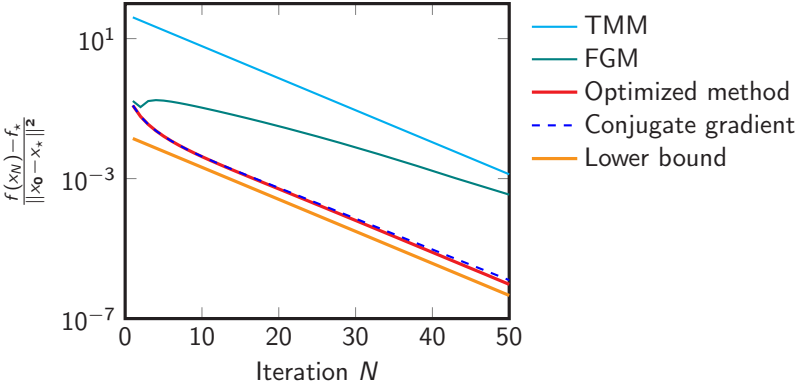

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- ◇ Beyond that, a few criterion/settings/methods for which "perfectly optimal" algorithms might be known, but matching lower bounds are still missing.
  - – $\frac{\|\nabla f(w_N)\|^2}{f(w_0)-f_\star}$,[11]
  - – a few numerically-generated methods, proximal variants, etc.

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Optimized gradient method (OGM)

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$$y_k = \frac{1}{\theta_{k,N}} z_k + \left(1 - \frac{1}{\theta_{k,N}}\right)\left(y_{k-1} - \frac{1}{L}\nabla f(y_{k-1})\right)$$

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$$\theta_{k+1,N} = \begin{cases} \frac{1+\sqrt{4\theta_{k,N}^2+1}}{2} & \text{if } k \leqslant N-2 \\ \frac{1+\sqrt{8\theta_{k,N}^2+1}}{2} & \text{if } k = N-1, \end{cases}$$

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where $\theta_{-1,N} = 0$, and roughly $\theta_{k,N} \approx \frac{k}{2}$.

The (tight) worst-case guarantee: $\frac{f(y_N)-f_\star}{L\|y_0-y_\star\|^2} \leqslant \frac{1}{2\theta_{N,N}^2} \approx \frac{2}{N^2}$. Matches exactly lower bound.[13]

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[13] Drori (2017). "The exact information-based complexity of smooth convex minimization." Journal of Complexity 39.

Information-Theoretic Exact Method (ITEM)

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$$y_k = (1-\beta_k)z_k + \beta_k\left(y_{k-1} - \frac{1}{L}\nabla f(y_{k-1})\right)$$

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$$A_{k+1} = \frac{(1+\frac{\mu}{L})A_k + 2\left(1+\sqrt{(1+A_k)(1+\frac{\mu}{L}A_k)}\right)}{(1-\frac{\mu}{L})^2}, \quad k \geqslant 0,$$

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with $A_0 = 0$. The (tight) guarantee is $\frac{\|z_N - z_\star\|^2}{\|z_0 - z_\star\|^2} \leqslant \frac{1}{1+\frac{\mu}{L}A_N} = O\left(\left(1-\sqrt{\frac{\mu}{L}}\right)^{2N}\right)$. Matches exact lower bound.[15]

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[15] Drori, T (2022). "On the oracle complexity of smooth strongly convex minimization." Journal of Complexity 68.

A few instructive examples

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... including "brutal" examples:

- Grimmer (2024). "Provably faster gradient descent via long steps." SIAM Journal on Optimization 34(3).
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Forewords and problem statement

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Concluding remarks

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